Constrained XY simulations

Transition curves and the 3D spin scaling dimension

Briefing for a collaborator's talk | Results attained 13-15 October 2026. This is an exploratory numerical report, not a precision universality determination. All quantities and figures below use the saved, independently reviewed analyses; no new MC was generated for this report.

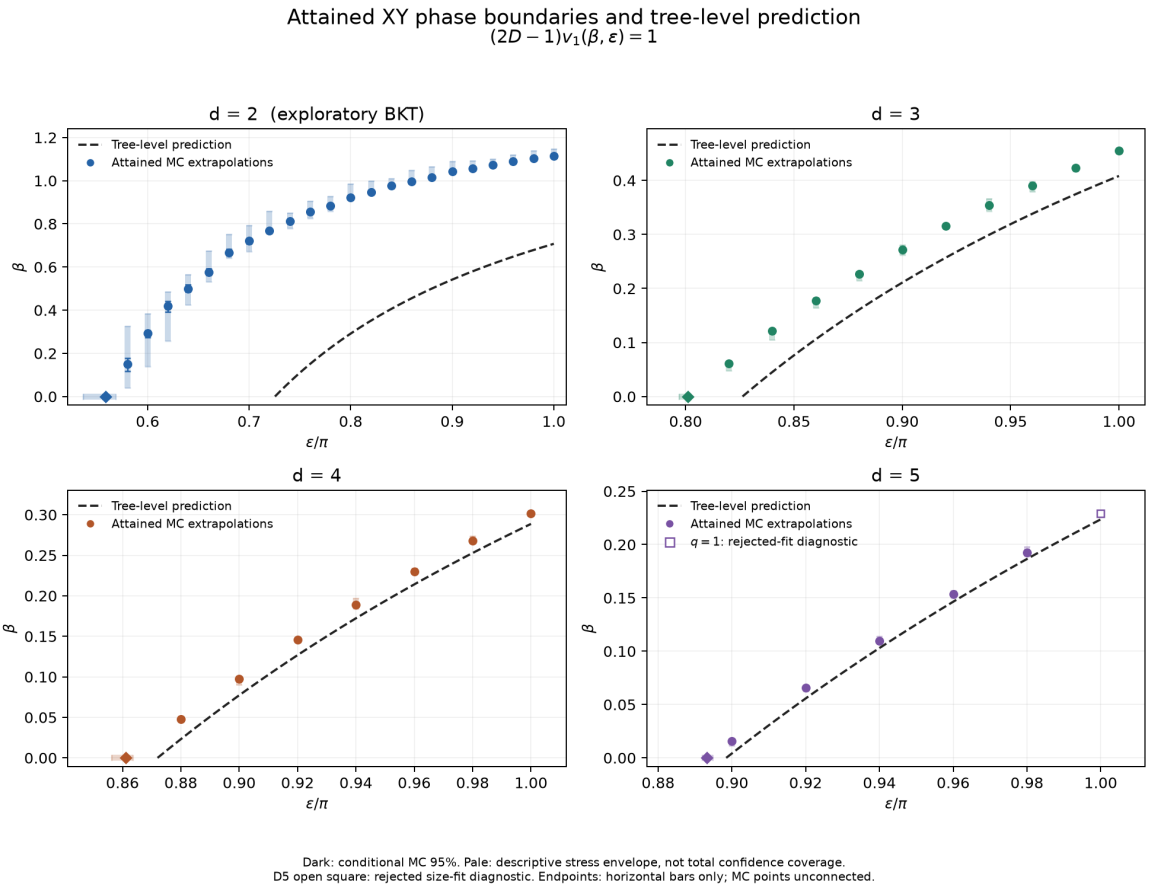


Figure 1. Fixed-ε points have vertical bars; β=0 endpoints have horizontal bars. Dark: conditional MC95. Pale: descriptive sensitivity unions, including MC noise and model choices; these are not calibrated total 95% intervals. The open 5D square at ε/π=1 is a rejected-fit diagnostic. Panel scales differ.

What these results support

The attained curves show a descriptive dimension-dependent approach toward the full tree-level prediction. The β=0 transition is at ε/π greater than 1/2 in every studied dimension. In 3D, common-correlation-length magnetization quotients give effective spin dimensions near **Δ ≈ 0.52**, compatible with the ordinary O(2) Wilson-Fisher expectation. A calibrated combined statistical-plus-systematic error has not been established.

Dimension	Positive-β estimates on the 0.02 ε/π grid	β=0 endpoint ε/π (approx.)
2	22: 0.58 through 1.00	0.559; broad model sensitivity
3	10: 0.82 through 1.00	0.801; finer crossing study on p. 3
4	7: 0.88 through 1.00	0.861; exploratory size fit
5	5 conditional: 0.90 through 0.98; 1.00 diagnostic only	0.893

Simulation and analysis methods

Model, boundaries and sampling

Angles φ on periodic hypercubic boxes of volume $V=L^d$. The dimensionless action is $S = -\beta \Sigma_{links} \cos \theta$, with $\theta = \mathrm{wrap}(\varphi_j-\varphi_i)$ and the strict constraint $|\theta| < \varepsilon$. Thus $\beta=0$ removes the cosine weight but preserves the hard constraint; $\varepsilon=\pi$ is the standard XY limit up to measure-zero configurations. L and N denote the same linear box size.

One recorded hybrid MC cycle = one local Metropolis sweep (one proposal per site), four Wolff cluster reflections, and one random-axis/sign unit-winding proposal. Four clusters are not four volume sweeps. Each chain has its own seed; fresh pairs generally start in zero and unit axial winding backgrounds. Measurements follow every cycle. Update/replay tests and full-field closure audits check implementation and recorded observables; they do not prove equilibration or independent sampling.

Study	Linear sizes L	Recorded cycles per chain (including initial transient)
2D transition curves	Reused 16,32,64 anchors; new 64,128,256; 512 at ε/π=0.60,0.80,1 and β=0	New positive-β cohort: 416 × 12,000. Endpoint chains separate; legacy data retain their own analysis cuts. Completed 1024 probes excluded.
3D transition curves	Source 6,8,12,16,24,32; primary fits start at 6; 32 diagnostic	788-source positive-curve analysis: 786 × 4,000 and 2 × 16,000. Includes diagnostic/β=0 coordinates.
4D transition curves	6,8,10,12,16,20,24; primary positive fits L≥10	728-source analysis: 726 × 4,000 and 2 × 16,000. Includes diagnostic/β=0 coordinates.
5D transition curves	Pilot 4; primary 6,8,10,12	282 trajectories: 208 × 4,000 and 74 × 8,000. Adaptive, not a balanced rectangular grid.
3D β=0 spin dimension	6,8,12,16,24,32,48,64; largest new chains 48^3 and 64^3	270 pinned sources: 186 × 12,000; 44 × 20,000; 24 × 8,000; 16 × 4,000. No box above 64^3.

Observables and finite-size transition definitions

Define $m = V^{-1}\Sigma \exp(i\varphi)$. The O(2) Binder convention is $U = 1 - \langle|m|^4\rangle/[2\langle|m|^2\rangle^2]$. The second-moment ratio is $R=\xi/L = \sqrt{(\langle S(0)\rangle/\langle F\rangle-1)/[2L \sin(\pi/L)]}$, with F averaged over axial minimal momenta. Ratios are formed from ensemble moments, not averaged configuration-by-configuration ratios. Vortex density, all axial winding histories, energy-like observables and update diagnostics are also recorded.

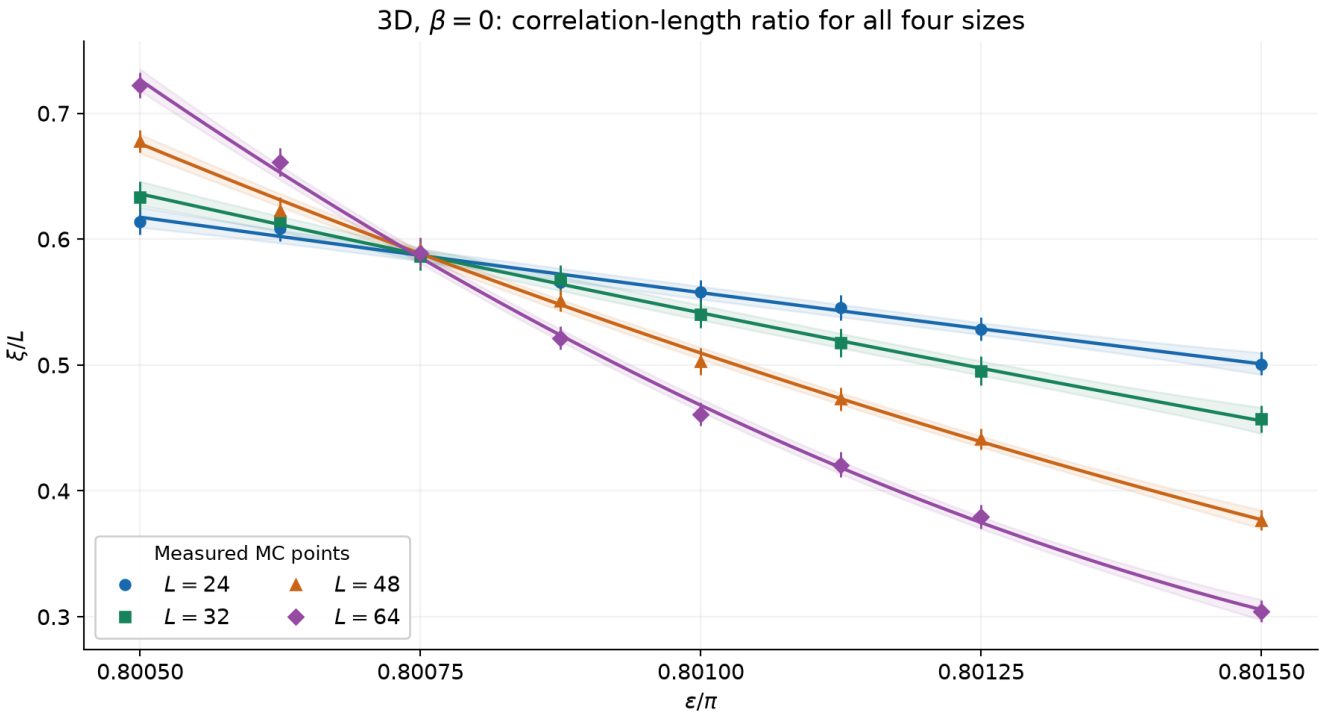
2D: local roots of $R\to0.7506912$ and $n_{eff}(L,2L)=-\ln[\langle|m|^2\rangle_{2L}/\langle|m|^2\rangle_L]/\ln 2 \to 1/4$; exploratory shifts proportional to $(\ln L+C)^{-2}$, shared infinite-volume limit and separate observable amplitudes. **3D/4D/5D:** local U=1/4 roots. Leading shift coordinates are respectively $L^{-1/0.67169}$, $L^{-2}(\ln L)^{-1/10}$, and $L^{-5/2}$. Size cuts and correction terms are tested, not assumed validated. In 5D the zero-mode Binder value is $1-\pi/4\approx0.2146$, so 1/4 is a shifted fixed criterion.

Statistical treatment

Typical primary cuts discard cycles 0-1000 (3D/4D/5D curves and spin study); new 2D/endpoint analyses generally cut through 2000. Retained histories are resampled in contiguous autocorrelation-aware blocks separately within starts, then pooled by retained rows. Original 3D/4D positive curves use 1,000 bootstrap draws; 2D, 5D and spin analyses use 2,000. Typical nominal block width is $\max(20, \mathrm{ceil}[10 \max \tau_{int}])$; the spin baseline doubles it and a further doubling is tested. Shared draws preserve covariance of derived observables and overlapping size pairs. Narrow intervals remain conditional on mixing, block choice, local windows and scaling laws.

3D at β=0: locating criticality

The target is the scaling dimension of the charge-one spin operator exp(iφ), not the magnetization critical exponent β_{mag}. In 3D, $\langle \exp(i\varphi(x))\, \exp(-i\varphi(0))\rangle \sim |x|^{-2\Delta}$, with $\Delta=(1+\eta)/2$. Correspondingly $\langle |m|^2\rangle \sim L^{-2\Delta}$, up to finite-size corrections and analytic backgrounds.



Four independent quadratic fits; no common crossing imposed.
Points and shaded bands: conditional MC 95%, excluding fit-model uncertainty.

Figure 2. All four large sizes on one axis. Each ξ/L response is fitted independently; neither a shared crossing nor a target critical R is imposed. Curves may have different finite-size crossings. The near-common intersection is useful evidence, not proof of an exact thermodynamic critical coupling.

Common-coupling quotient method

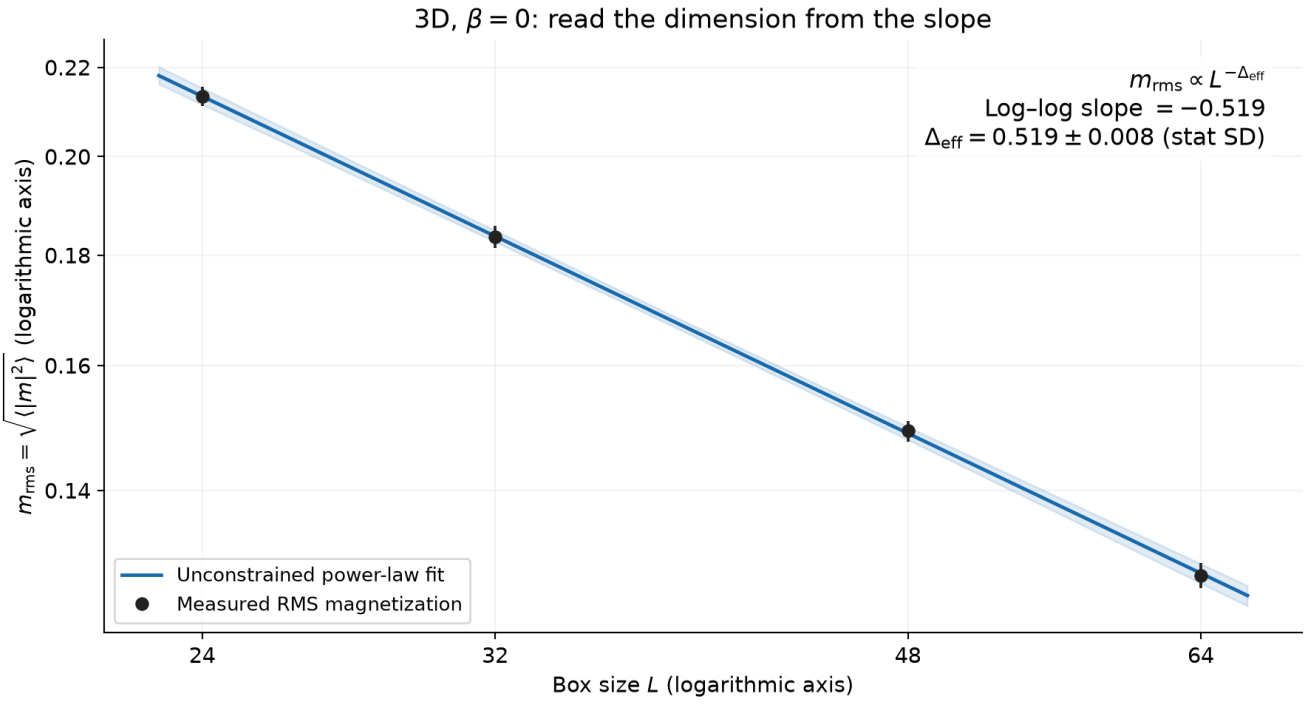
For each pair L:2L, solve $R_L(q^*)=R_{2L}(q^*)$ at the **same** q=ε/π. Then evaluate $\Delta_{eff}=-\ln[\langle |m|^2\rangle_{2L,q^*}/\langle |m|^2\rangle_{L,q^*}]/(2\ln 2)$. This avoids matching magnetizations at different ε or imposing a universal Binder/R value. The final 120 trajectories refine common knots with fine interior spacing 0.000125; L=48,64 chains have 20,000 cycles.

Noisy knot-by-knot interpolation can yield multiple crossings. The displayed results instead use weighted local quadratic responses for R and ⟨|m|²⟩ over q=0.800500-0.801500, with original joint bootstrap draws. Inner-window, linear, cut, block and start-parity alternatives are retained. A required root must be physical, unique in its window and present in at least 99% of bootstrap draws.

Finite size pair	Crossing q*	Δ_eff ± statistical SD
24:48	0.8007565	0.51598 ± 0.00292
32:64	0.8007428	0.51903 ± 0.00326

Both pairs retain 2,000/2,000 roots for each of the five broad-quadratic baseline/cut/block/start variants. They are correlated, not two independent estimates to average. Inner-quadratic 24:48 retains only 97% and remains unresolved; broad-linear fits fail the large-size curvature check. Thus the selected-family coverage does not validate all response choices.

Reading Δ and answering questions



Fixed measured $\varepsilon/\pi = 0.80075$, near the ξ/L crossings; four sizes fitted together.
Bars/band: conditional MC 95%. Coupling-location and finite-size corrections are not included in this fit error.

Figure 3. A direct slope visualization at exactly measured q=0.80075: $\sqrt{\langle |m|^2 \rangle} \sim L^{-\Delta_eff}$. Correlated fit over L=24,32,48,64 gives **Δ_eff=0.51942 ± 0.00795 stat** (one SD); conditional MC95 [0.50398,0.53505]. This fixed-q near-critical slope excludes uncertainty in thermodynamic criticality and scaling corrections.

How to quote the result

Suggested talk wording: "At β=0, finite-size magnetization quotients and a direct magnetization slope give Δ near 0.52, compatible with the standard 3D O(2) Wilson-Fisher CFT. The statistical precision is a few 10^{-3} for selected pair quotients, but finite-size and response-model uncertainty are not yet calibrated." Do not turn this into Δ=0.52±0.01 as a formal confidence interval or quote an invented σ_{sys}.

Conditional correlated extrapolations with a fixed Wilson-Fisher-informed correction exponent ω=0.789 give 0.52321±0.00346 stat (minimum pair L=8) and 0.51834±0.00616 (minimum L=12); an inner-linear alternative gives 0.52057±0.00640. These are sensitivity examples, not a selected final CFT value. The raw unsmoothed 32:64 quotient is 0.52593±0.00572, with start-parity sensitivity; local regression changes the estimator. Literature comparison: Δ=0.519088(22) from conformal bootstrap [1], which also tabulates the MC result 0.519050(40). Neither was imposed as the measured Δ.

Questions and numerical context

Qualifications worth saying aloud

What do the pale bars mean? A descriptive union across tested cuts, thresholds, interpolants and scaling models, including amplified MC noise and some rejected-model diagnostics; not a systematic standard deviation or total confidence interval.
What remains unresolved? Absolute BKT accuracy: the 2D q=1 dark interval misses the known 1.1199 benchmark [2], although its wider union includes it. The 5D q=1 leading size fit rejects and is diagnostic only. Low independent-block counts, including as few as seven in some 5D starts, and start/half-history flags remain material.
Why can order survive at β=0? The angle constraint survives heating and can create entropic rigidity. Vortices being permitted at ε>π/2 does not imply their macroscopic proliferation. In 2D say “massless/BKT phase,” not spontaneous magnetization.

Attained β=0 endpoints

d	ε_c/π	Conditional MC95	Descriptive sensitivity union
2	0.55857	[0.55590, 0.56174]	[0.53629, 0.56855]
3	0.80108	[0.80076, 0.80144]	[0.79704, 0.80192]
4	0.86112	[0.86093, 0.86136]	[0.85622, 0.86346]
5	0.89320	[0.89303, 0.89332]	[0.89191, 0.89476]

These are the fixed phase-delivery endpoint extrapolations, not the later finite-pair common-ξ crossings in 3D. The 4D leading endpoint fit has p≈0.036 and remains exploratory; all sensitivity unions are descriptive.

References and reproducibility

[1] Chester et al., JHEP 06 (2020) 142, arXiv:1912.03324, Table 1 / Eq. 40. [2] Bietenholz, Gerber & Rejón-Barrera, JSTAT (2013) P12009, arXiv:1307.0485. [3] Above-upper-critical-dimension background: Chen-Dohm, cond-mat/9809394; Wittmann-Young, arXiv:1410.5296. Tree comparison: supplied constrained_xy_large_d_note.pdf, pp. 2 and 5: (2d-1)v1=1, v1=∫ exp(β cosθ)cosθ dθ / ∫ exp(β cosθ)dθ on [-ε,ε]. It is a tree instability approximation, not an exact hypercubic curve.

Companion bundle: 13 figures in PNG/PDF/SVG, a captioned atlas, numerical tables, captions and SHA-256 provenance. Source snapshots: phase delivery 20261014-182254; local-xi response 20261015-213702; dimension readout 20261015-224446. All observable/RNG/winding histories, exact checkpoints and logs are preserved; field-only retention does not remove the observable series used here.